

# Annex B

## Cover Page for College to submit with Prelim Examination

### Paper

From MJC :

| Qn/No | Topic Set                         | Paper Number | Answers                                                                                                                                                      | Marks |
|-------|-----------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1     | Functions                         | 9740/01      | (ii) $D_g = (1, \infty), R_{fg} = [-1, 1]$                                                                                                                   | 5     |
| 2     | Complex Numbers                   | 9740/01      | (a) $z = i, w = 1 - 2i$<br>(b) $k = \sqrt{3}$                                                                                                                | 5     |
| 3     | Integration Techniques            | 9740/01      | (i) $\frac{1}{4} \ln(1 + 2x^2) + \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2}x) + C$<br>(ii) $\frac{x^3}{3} (\ln x)^2 - \frac{2x^3}{9} \ln x + \frac{2x^3}{27} + C$ | 6     |
| 4     | Mathematical Induction            | 9740/01      |                                                                                                                                                              | 5     |
| 5     | Curve Sketching & Transformations | 9740/01      |                                                                                                                                                              | 7     |
| 6     | Vectors                           | 9740/01      | (b) $p = k(3q - 2r), k \in \mathbb{R}$                                                                                                                       | 6     |
| 7     | Complex Numbers                   | 9740/01      | $z = 0 + 0i,$<br>$z = 2 + 2i$                                                                                                                                | 7     |
| 8     | Recurrence Relations [Analysis]   | 9740/01      | (a)(i)<br>$x_n = 0.9x_{n-1} + 90, n = 1, 2, 3, \dots$<br>(a)(iii) $x_n \rightarrow 900$ as $n \rightarrow \infty$<br>(b) $a = 90$                            | 8     |
| 9     | Complex Numbers                   | 9740/01      | (iii)<br>$z = \sqrt{3} - i, -\sqrt{3} - i, -\frac{\sqrt{3}}{3} - i, -i$                                                                                      | 9     |
| 10    | Integration Applications          | 9740/01      | (i) $\ln 2 - \frac{1}{2}$<br>(ii) $\frac{\pi^2}{2} - \frac{4\pi}{3}$                                                                                         | 10    |

|    |                               |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
|----|-------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 11 | Maclaurin's Expansions        | 9740/01 | <p>(a) <math>\sum_{r=1}^{\infty} \frac{1}{r!} = e - 1</math></p> <p>(b)(i) <math>y \approx 2x - \frac{8}{3}x^3</math></p> <p>(b)(ii) <math>\frac{73}{1875}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 |
| 12 | Vectors                       | 9740/01 | <p>(i)</p> $\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}, \lambda \in \mathbb{R}$ <p>or</p> $\mathbf{r} = \begin{pmatrix} \frac{29}{3} \\ \frac{2}{3} \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}, \lambda \in \mathbb{R}$ <p>(ii) Position vector = <math>\frac{1}{3} \begin{pmatrix} 10 \\ 5 \\ -11 \end{pmatrix}</math></p> <p>and,</p> <p>Length of projection = <math>\frac{\sqrt{3}}{3}</math></p> <p>(iii)</p> $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -29 \\ -13 \end{pmatrix} = 0 \quad \text{or} \quad \mathbf{r} \cdot \begin{pmatrix} -2 \\ 29 \\ 13 \end{pmatrix} = 0$ | 12 |
| 13 | Differentiation & Application | 9740/01 | <p>(a) <math>\frac{2}{\sqrt{1-4x^2}}</math>, (for <math>x \geq 0</math>)</p> <p>and,</p> $\frac{-2}{\sqrt{1-4x^2}}, \text{ (for } x < 0 \text{)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 10 |

|    | Pure Maths                      |         |                                                                                                                                 |    |
|----|---------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------|----|
| 1  | Inequalities                    | 9740/02 | $-\frac{5}{4} < x < 1$                                                                                                          | 4  |
| 2  | AP & GP                         | 9740/02 | $a = \{a \in \mathbb{R} : 2 < a < 3\}$<br>and, Least $n = 31$                                                                   | 6  |
| 3  | Summation                       | 9740/02 | $\sum_{n=3}^N \frac{1}{n(n-1)(n-2)}$ $= \frac{1}{4} - \frac{1}{2N(N-1)}$                                                        | 8  |
| 4  | Binomial Expansions             | 9740/02 | (a) $(1+x)^8 =$<br>$1+8x+28x^2+56x^3+\dots$<br>and, $k = -1$<br><br>(b) $\frac{3}{2} - \frac{5x}{4} + \frac{9x^2}{8} + \dots$   | 10 |
| 5  | Differential Equation           | 9740/02 | (a) $\frac{dz}{dx} = x \frac{d^2y}{dx^2} + \frac{dy}{dx}$<br><br>and,<br>$y = \frac{x^3}{3} + c \ln x + d, c, d \in \mathbb{R}$ | 12 |
|    | Statistics                      |         |                                                                                                                                 |    |
| 6  | P & C                           | 9740/02 | (a) 15400<br>(b) 120960                                                                                                         | 4  |
| 7  | Sampling Method                 | 9740/02 |                                                                                                                                 | 5  |
| 8  | Binomial                        | 9740/02 | (i) 6, $P(X = 5) = 0.172$<br>(ii) Least $n=21$                                                                                  | 6  |
| 9  | Estimation & Hypothesis Testing | 9740/02 | (i) $\bar{x} = 1.96, s^2 = 0.01$<br>(ii) $\{\alpha \in \mathbb{R} : 4.48 \leq \alpha \leq 100\}$                                | 7  |
| 10 | Probability                     | 9740/02 | (iii) $P(A'   B') = \frac{3}{4}$<br>Required probability<br>$= 0.00405$                                                         | 8  |

|    |                          |         |                                                              |    |
|----|--------------------------|---------|--------------------------------------------------------------|----|
| 11 | Poisson                  | 9740/02 | (i) 0.594<br>(ii) 0.857 , 0.406 , 6                          | 9  |
| 12 | Normal Distribution      | 9740/02 | (i) 0.131<br>(ii) 0.967<br>(iii) 0.0931                      | 11 |
| 13 | Regression & Correlation | 9740/02 | (ii) $h = 217 - 3.09s$<br>(v) $s = 29.9 \approx 30$ revs/min | 10 |

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|---------|---------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------|-------|
|         | <b>Pure Maths</b>                     |              |                                                                                                                         |       |
| 1       | Simultaneous Equation                 | 8863/01      | $a = -2, b = 2$ (rejected $a > 0$ )<br>or, $a = \frac{9}{4}, b = -\frac{1}{8}$                                          | 5     |
| 2       | Functions                             | 8863/01      | (ii) $D_g = (1, \infty)$ ,<br>$fg(x) = \cos(\ln(x^2))$ ,<br>$R_{fg} = [-1, 1]$                                          | 6     |
| 3       | Curve Sketching                       | 8863/01      | (i) $x = \frac{1}{2}, y = -\frac{1}{2}$<br>(ii) No. of roots = 3                                                        | 9     |
| 4       | Integration + Applications            | 8863/01      | (i) intersection pts:<br>(2,9) and (3,4) for $x \geq 0$<br>(ii) $13x - \frac{x^3}{3} + C$<br>(iii) Area = $\frac{2}{3}$ | 10    |
| 5       | Differentiation & Applications        | 8863/01      | (a) (i) $y = 54x - 322$<br>(ii) $y = -\frac{1}{54}x + \frac{19}{18}$<br>(b) $x = \frac{40}{3\pi + 8}$                   | 10    |
|         | <b>Statistics</b>                     |              |                                                                                                                         |       |
| 6       | Sampling Methods                      | 8863/01      | (i) Systematic sampling                                                                                                 | 6     |
| 7       | Probability                           | 8863/01      | (i) $p = 2$<br>(ii) 0.3465<br>(iii) 0.354, 0.817                                                                        | 10    |
| 8       | Hypothesis Testing & Estimation       | 8863/01      | (i) $\mu = (102.4085, 1080515)$                                                                                         | 10    |
| 9       | Binomial Distribution & Approximation | 8863/01      | (i) $E(X) = 1$<br>(ii) $P(X \geq 2) = 0.264$<br>(iii) $P(X \leq 4   x \geq 2) = 0.988$<br>$P(Y \geq 20) = 0.909$        | 10    |
| 10      | Regression + Correlation              | 8863/01      | (i) $r = 0.155$<br>(iv)(a) $y = 39.945 + 0.898x$<br>(iv)(b) $y = -7.20 + 0.907x$                                        | 12    |

|    |                           |         |                                                               |    |
|----|---------------------------|---------|---------------------------------------------------------------|----|
|    |                           |         | (v)(a) 159<br>(v)(b) 271                                      |    |
| 11 | Normal Distribution & CLT | 8863/01 | (i) 0.0912<br>(ii) 0.932<br>(iii) Least $n = 4$<br>(iv) 0.987 | 12 |